CHAPTER 29

WIND LOADS ON BUILDING APPURTENANCES AND OTHER STRUCTURES: MAIN WIND FORCE RESISTING SYSTEM (DIRECTIONAL PROCEDURE)

29.1 SCOPE

29.1.1 Structure Types. This chapter applies to the determination of wind loads on building appurtenances (such as rooftop structures and rooftop equipment) and other structures of all heights (such as solid freestanding walls and freestanding solid signs, chimneys, tanks, open signs, single-plane open frames, and trussed towers) using the Directional Procedure.

The steps required for the determination of wind loads on building appurtenances and other structures are shown in Table 29.1-1. The steps required to determine wind loads on main wind force resisting system (MWFRS) on circular bins, silos, and tanks are in Table 29.1-2.

> **User Note:** Use Chapter 29 to determine wind pressures on the MWFRS of solid freestanding walls, freestanding solid signs, chimneys, tanks, open signs, single-plane open frames, and trussed towers. Wind loads on rooftop structures and equipment may be determined from the provisions of this chapter. The wind pressures are calculated using specific equations based upon the Directional Procedure.

Table 29.1-1 Steps to Determine Wind Loads on MWFRS Rooftop Equipment and Other Structures

Step 1: Determine Risk Category of building or other structure; see Table 1.5-1.
Step 2: Determine the basic wind speed, V, for applicable Risk Category; see Figs. 26.5-1 and 26.5-2.
Step 3: Determine wind load parameters:
- Wind directionality factor, K_d; see Section 26.6 and Table 26.6-1.
- Exposure category B, C, or D; see Section 26.7.
- Topographic factor, K_{zt}; see Section 26.8 and Fig. 26.8-1.
- Ground elevation factor, K_e; see Section 26.9 and Table 26.9-1
- Gust-effect factor, G; see Section 26.11, except for rooftop equipment.
- Combined (GC_r) factor for rooftop equipment; see Section 29.4.1.

Step 4: Determine velocity pressure exposure coefficient, K_z or K_h; see Table 26.10-1.
Step 5: Determine velocity pressure q_z or q_h; see Eq. (26.10-1).
Step 6: Determine force coefficient, C_f, except for rooftop equipment:
- Solid freestanding signs or solid freestanding walls, Fig. 29.3-1.
- Chimneys, tanks, Fig. 29.4-1.
- Open signs, single-plane open frames, Fig. 29.4-2.
- Trussed towers, Fig. 29.4-3.
- Rooftop equipment, using combined (GC_r) factors listed in Section 29.4.1.
- Rooftop solar panels, Fig. 29.4-7 and Eq. (29.4-6), or Fig. 29.4-8.

Step 7: Calculate wind force, F, or pressure, p:
- Eq. (29.3-1) for signs and walls.
- Eqs. (29.4-2) and (29.4-3) for rooftop structures and equipment.
- Eq. (29.4-1) for other structures.
- Eq. (29.4-5) or (29.4-7) for rooftop solar panels.

29.1.2 Conditions. An appurtenance or structure that has design wind loads determined in accordance with this section shall comply with all of the following conditions:

1. The structure is a regular-shaped structure as defined in Section 26.2; and
2. The structure does not have response characteristics making it subject to across-wind loading, vortex shedding, or instability caused by galloping or flutter; nor does it have a site location for which channeling effects or buffeting in the wake of upwind obstructions warrant special consideration.

29.1.3 Limitations. The provisions of this chapter take into consideration the load magnification effect caused by gusts in resonance with along-wind vibrations of flexible structures. Structures that do not meet the requirements of Section 29.1.2 or that have unusual shapes or response characteristics shall be designed using recognized literature documenting such wind load effects or shall use the Wind Tunnel Procedure specified in Chapter 31.

29.1.4 Shielding. There shall be no reductions in velocity pressure caused by apparent shielding afforded by buildings and other structures or terrain features.

Table 29.1-2 Steps to Determine Wind Loads on MWFRS Circular Bins, Silos, and Tanks

Step 1: Determine Risk Category of structure; see Table 1.5-1.
Step 2: Determine the basic wind speed, V, for applicable Risk Category; see Figs. 26.5-1 and 26.5-2.
Step 3: Determine wind load parameters:
- Wind directionality factor, K_d; see Section 26.6 and Table 26.6-1.
- Exposure category B, C, or D; see Section 26.7.
- Topographic factor, K_{zt}; see Section 26.8 and Fig. 26.8-1.
- Ground elevation factor, K_e; see Section 26.9 and Table 26.9-1
- Enclosure classification, see Section 26.12.
- Internal pressure coefficient, (GC_{pi}), see Table 26.13-1.
- Gust-effect factor, G; see Section 26.11.

Step 4: Determine velocity pressure exposure coefficient, K_z or K_h; see Table 26.10-1.
Step 5: Determine velocity pressure q_h; see Eq. (26.10-1).
Step 6: Determine force coefficient for walls, see Sections 29.4.2.1 and 29.4.2.4.
Step 7: Determine external pressure coefficient (GC_p) for roofs and undersides if elevated, see Sections 29.4.2.2 and 29.4.2.3.
Step 8: Calculate wind force, F, or pressure, p:
- Eq. (29.4-1) for walls.
- Eq. (29.4-4) for roofs.

29.2 GENERAL REQUIREMENTS

29.2.1 Wind Load Parameters Specified in Chapter 26. The following wind load parameters shall be determined in accordance with Chapter 26:

- Basic wind speed, V (Section 26.5);
- Wind directionality factor, K_d (Section 26.6);
- Exposure category (Section 26.7);
- Topographic factor, K_{zt} (Section 26.8);
- Ground elevation factor, K_e (Section 26.9); and
- Enclosure classification (Section 26.12).

29.3 DESIGN WIND LOADS: SOLID FREESTANDING WALLS AND SOLID SIGNS

29.3.1 Solid Freestanding Walls and Solid Freestanding Signs. The design wind force for solid freestanding walls and solid freestanding signs shall be determined by the following formula:

$$F = q_h G C_f A_s \text{ (lb)} \qquad (29.3\text{-}1)$$

$$F = q_h G C_f A_s \text{ (N)} \qquad (29.3\text{-}1.\text{si})$$

where

q_h = velocity pressure evaluated at height h (defined in Fig. 29.3-1) as determined in accordance with Section 26.10.
G = gust-effect factor from Section 26.11.
C_f = net force coefficient from Fig. 29.3-1.
A_s = gross area of the solid freestanding wall or freestanding solid sign, in ft^2 (m^2).

29.3.2 Solid Attached Signs. The design wind pressure on a solid sign attached to the wall of a building, where the plane of the sign is parallel to and in contact with the plane of the wall, and the sign does not extend beyond the side or top edges of the wall, shall be determined using procedures for wind pressures on walls in accordance with Chapter 30 and setting the internal pressure coefficient (GC_{pi}) equal to 0.

This procedure shall also be applicable to solid signs attached to but not in direct contact with the wall, provided that the gap between the sign and wall is no more than 3 ft (0.9 m) and the edge of the sign is at least 3 ft (0.9 m) in from free edges of the wall, i.e., side and top edges and bottom edges of elevated walls.

29.4 DESIGN WIND LOADS: OTHER STRUCTURES

The design wind force for other structures (chimneys, tanks, open signs, single-plane open frames, and trussed towers), whether ground or roof mounted, shall be determined by the following equation:

$$F = q_z G C_f A_f \text{ (lb)} \qquad (29.4\text{-}1)$$

$$F = q_z G C_f A_f \text{ (N)} \qquad (29.4\text{-}1.\text{si})$$

where

q_z = velocity pressure evaluated at height z, as defined in Section 26.10, of the centroid of area A_f.
G = gust-effect factor from Section 26.11.
C_f = force coefficients from Figs. 29.4-1 through 29.4-4.
A_f = projected area normal to the wind except where C_f is specified for the actual surface area, in ft^2 (m^2).

Guidance for determining G, C_f, and A_f for structures found in petrochemical and other industrial facilities that are not otherwise addressed in ASCE 7 can be found in the *Wind Loads for Petrochemical and Other Industrial Facilities* (2011), published by ASCE, Reston, VA.

29.4.1 Rooftop Structures and Equipment for Buildings. The lateral force, F_h, and vertical force, F_v, for rooftop structures and equipment, except as otherwise specified for roof-mounted solar panels (Sections 29.4.3 and 29.4.4) and structures identified in Section 29.4, shall be determined as specified following.

The resultant lateral force, F_h, shall be determined from Eq. (29.4-2) and applied at a height above the roof surface equal to or greater than the centroid of the projected area, A_f.

$$F_h = q_h (GC_r) A_f \text{ (lb)} \qquad (29.4\text{-}2)$$

$$F_h = q_h (GC_r) A_f \text{ (N)} \qquad (29.4\text{-}2.\text{si})$$

where

(GC_r) = 1.9 for rooftop structures and equipment with A_f less than $(0.1Bh)$. (GC_r) shall be permitted to be reduced linearly from 1.9 to 1.0 as the value of A_f is increased from $(0.1Bh)$ to (Bh).
q_h = velocity pressure evaluated at mean roof height of the building.
A_f = vertical projected area of the rooftop structure or equipment on a plane normal to the direction of wind, in ft^2 (m^2).

The vertical uplift force, F_v, on rooftop structures and equipment shall be determined from Eq. (29.4-3):

$$F_v = q_h (GC_r) A_r \text{ (lb)} \qquad (29.4\text{-}3)$$

$$F_v = q_h (GC_r) A_r \text{ (N)} \qquad (29.4\text{-}3.\text{si})$$

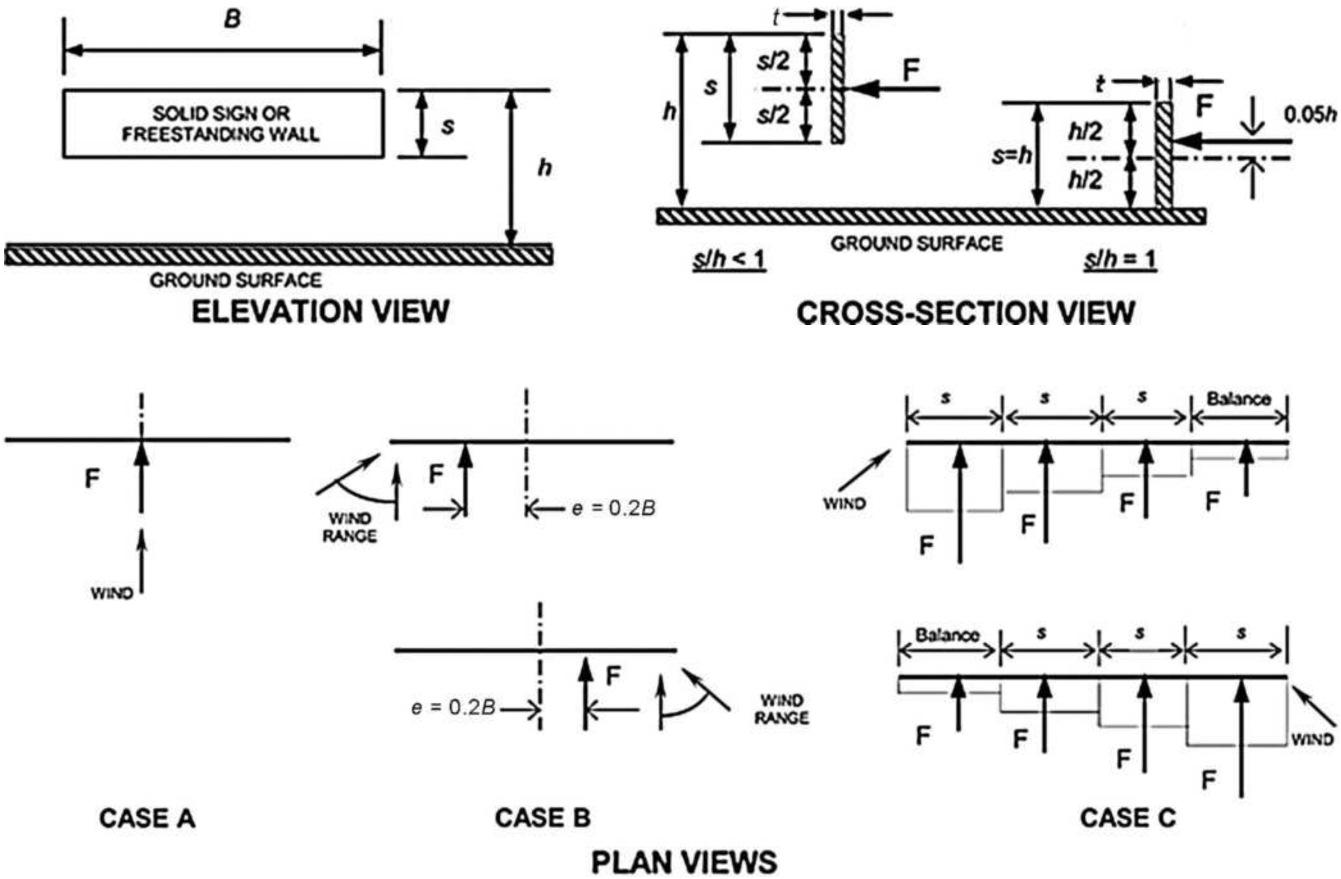
where

(GC_r) = 1.5 for rooftop structures and equipment with A_r less than $(0.1BL)$. (GC_r) shall be permitted to be reduced linearly from 1.5 to 1.0 as the value of A_r is increased from $(0.1BL)$ to (BL).
q_h = velocity pressure evaluated at the mean roof height of the building.
A_r = horizontal projected area of rooftop structure or equipment, in ft^2 (m^2).

29.4.2 Design Wind Loads: Circular Bins, Silos, and Tanks with $h \le 120$ ft ($h \le 36.5$ m), $D \le 120$ ft ($D \le 36.5$ m), and $0.25 \le H/D \le 4$. Grouped circular bins, silos, and tanks of similar size with center-to-center spacing greater than two diameters shall be treated as isolated structures. For spacings less than 1.25 diameters, the structures shall be treated as grouped and the wind pressure shall be determined from Section 29.4.2.4. For intermediate spacings, linear interpolation of the C_p (or C_f) values shall be used.

29.4.2.1 External Walls of Isolated Circular Bins, Silos, and Tanks. To determine the overall drag on circular bins, silos, and tanks using Eq. (29.4-1), a drag coefficient (C_f) of 0.63 based on projected walls (DH) is permitted to be used, where H/D is in the range of 0.25 to 4.0 and the cylinder (diameter D) is standing on the ground or supported by columns. The clearance height (C) shall be less than or equal to the height of the solid cylinder (H) as shown in Fig. 29.4-4.

Diagrams



Notation

B = Horizontal dimension of sign, in ft (m)
e = Eccentricity of force, in ft (m)
F = Design wind force for other structures, in lb (N)
h = Height of the sign, in ft (m)
L_r = Horizontal dimension of return corner, in ft (m)
$R_{\min}$ = $t/$min (B and s)
$R_{\max}$ = $t/$max (B and s)
s = Vertical dimension of the sign, in ft (m)
t = Thickness of the sign in ft (m)
ε = Ratio of solid area to gross area

Force Coefficients, C_f, for Case A and Case B

Clearance Ratio, s/h	Aspect Ratio, B/s											
	≤ 0.05	0.1	0.2	0.5	1	2	4	5	10	20	30	≥45
1	1.80	1.70	1.65	1.55	1.45	1.40	1.35	1.35	1.30	1.30	1.30	1.30
0.9	1.85	1.75	1.70	1.60	1.55	1.50	1.45	1.45	1.40	1.40	1.40	1.40
0.7	1.90	1.85	1.75	1.70	1.65	1.60	1.60	1.55	1.55	1.55	1.55	1.55
0.5	1.95	1.85	1.80	1.75	1.75	1.70	1.70	1.70	1.70	1.70	1.70	1.75
0.3	1.95	1.90	1.85	1.80	1.80	1.80	1.80	1.80	1.80	1.85	1.85	1.85
0.2	1.95	1.90	1.85	1.80	1.80	1.80	1.80	1.80	1.85	1.90	1.90	1.95
≤ 0.16	1.95	1.90	1.85	1.85	1.80	1.80	1.85	1.85	1.85	1.90	1.90	1.95

FIGURE 29.3-1 Design Wind Loads (All Heights): Force Coefficients, C_f, for Other Structures—Solid Freestanding Walls and Solid Freestanding Signs

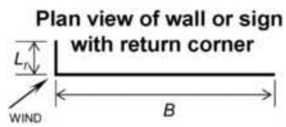
continues

Force Coefficients, C_f, for Case C

Region (horizontal distance from windward edge)	Aspect Ratio, B/s										
	2	3	4	5	6	7	8	9	10	13	≥45
0 to s	2.25	2.60	2.90	3.10*	3.30*	3.40*	3.55*	3.65*	3.75*	4.00*	4.30*
s to $2s$	1.50	1.70	1.90	2.00	2.15	2.25	2.30	2.35	2.45	2.60	2.55
$2s$ to $3s$		1.15	1.30	1.45	1.55	1.65	1.70	1.75	1.85	2.00	1.95
$3s$ to $10s$			1.10	1.05	1.05	1.05	1.05	1.00	0.95		
$3s$ to $4s$										1.50	1.85
$4s$ to $5s$										1.35	1.85
$5s$ to $10s$										0.90	1.10
$>10s$										0.55	0.55

*Values shall be multiplied by the following reduction factor when a return corner is present:

L_r/S	*Reduction Factor*
0.3	0.90
1.0	0.75
≥2	0.60



Notes

1. The term “signs” in these notes also applies to freestanding walls.
2. Signs with openings comprising less than 30% of the gross area are classified as solid signs. Force coefficients for solid signs with openings shall be permitted to be multiplied by the reduction factor $(1-(1-\varepsilon)^{1.5})$.
3. To allow for both normal and oblique wind directions, the following cases shall be considered:
   For $s/h < 1$:
   Case A: Resultant force acts normal to the face of the sign through the geometric center.
   Case B: Resultant force acts normal to the face of the sign at a distance from the geometric center toward the windward edge equal to 0.2 times the average width of the sign.
   For double-faced signs with all sides enclosed and $R_{max} \leq 0.4$, it is permitted to use force eccentricity, $e = (0.2 - 0.25R_{max})B$.
   For double-faced signs with all sides enclosed and $R_{min} \leq 0.75$, it is permitted to multiply tabulated C_f values in Cases A and B by the reduction factor, $(1 - 0.133R_{min})$.
   For $B/s \geq 2$, Case C must also be considered;
   Case C: Resultant forces act normal to the face of the sign through the geometric centers of each region.
   For $s/h = 1$:
   The same cases as above except that the vertical locations of the resultant forces occur at a distance above the geometric center equal to 0.05 times the average height of the sign.
4. For Case C where $s/h > 0.8$, force coefficients shall be multiplied by the reduction factor $(1.8 - s/h)$. It is permitted to apply this reduction with those specified in Note 3.
5. Linear interpolation is permitted for values of s/h, B/s, and L_r/s other than shown.

FIGURE 29.3-1 (*Continued*). Design Wind Loads (All Heights): Force Coefficients, C_f, for Other Structures—Solid Freestanding Walls and Solid Freestanding Signs

Force Coefficients, C_f

Cross Section	Type of Surface	h/D: 1	7	25
Square (wind normal to face)	All	1.3	1.4	2.0
Square (wind along diagonal)	All	1.0	1.1	1.5
Hexagonal or octagonal	All	1.0	1.2	1.4
Round, $D\sqrt{q_z} > 2.5$	Moderately smooth	0.5	0.6	0.7
$D\sqrt{q_z} > 5.3$ (in S.I.)	Rough ($D'/D = 0.02$)	0.7	0.8	0.9
	Very rough ($D'/D = 0.08$)	0.8	1.0	1.2
Round, $D\sqrt{q_z} \leq 2.5$ $D\sqrt{q_z} \leq 5.3$ (in S.I.)	All	0.7	0.8	1.2

Notation

D = Diameter of circular cross section and least horizontal dimension of square, hexagonal, or octagonal cross sections at elevation under consideration, in ft (m)
D' = Depth of protruding elements such as ribs and spoilers, in ft (m)
h = Height of structure, in ft (m)
q_z = Velocity pressure evaluated at height z above ground, in lb/ft^2 (N/m^2).

Notes

1. The design wind force shall be calculated based on the area of the structure projected on a vertical plane normal to the wind direction.
   The force shall be assumed to act parallel to the wind direction.
2. Linear interpolation is permitted for h/D values other than shown.

FIGURE 29.4-1 Other Structures (All Heights): Force Coefficients, C_f, for Chimneys, Tanks, and Similar Structures

Force Coefficients, C_f

		Rounded Members	
ε	Flat-Sided Members	$D\sqrt{q_z} \leq 2.5$ ($D\sqrt{q_z} \leq 5.3$) s.i	$D\sqrt{q_z} > 2.5$ ($D\sqrt{q_z} > 5.3$) s.i
<0.1	2.0	1.2	0.8
0.1 to 0.29	1.8	1.3	0.9
0.3 to 0.7	1.6	1.5	1.1

Notation

ε = Ratio of solid area to gross area
D = Diameter of a typical round member, in ft (m)
q_z = Velocity pressure evaluated at height z above ground, in lb/ft^2 (N/m^2)

Notes

1. Signs with openings making up 30% or more of the gross area are classified as open signs.
2. Calculation of the design wind forces shall be based on the area of all exposed members and elements projected on a plane normal to the wind direction. Forces shall be assumed to act parallel to the wind direction.
3. Area A_f consistent with these force coefficients is the solid area projected normal to the wind direction.

FIGURE 29.4-2 Other Structures (All Heights): Force Coefficients, C_f, for Open Signs and Single-Plane Open Frames

Force Coefficients, C_f

Tower Cross Section	C_f
Square	$4.0\varepsilon^2 - 5.9\varepsilon + 4.0$
Triangle	$3.4\varepsilon^2 - 4.7\varepsilon + 3.4$

Notation

ε = Ratio of solid area to gross area of one tower face for the segment under consideration.

Notes

1. For all wind directions considered, the area A_f consistent with the specified force coefficients shall be the solid area of a tower face projected on the plane of that face for the tower segment under consideration.
2. The specified force coefficients are for towers with structural angles or similar flat-sided members.
3. For towers containing rounded members, it is acceptable to multiply the specified force coefficients by the following factor when determining wind forces on such members:

$$0.51\varepsilon^2 + 0.57, \text{but not} > 1.0$$

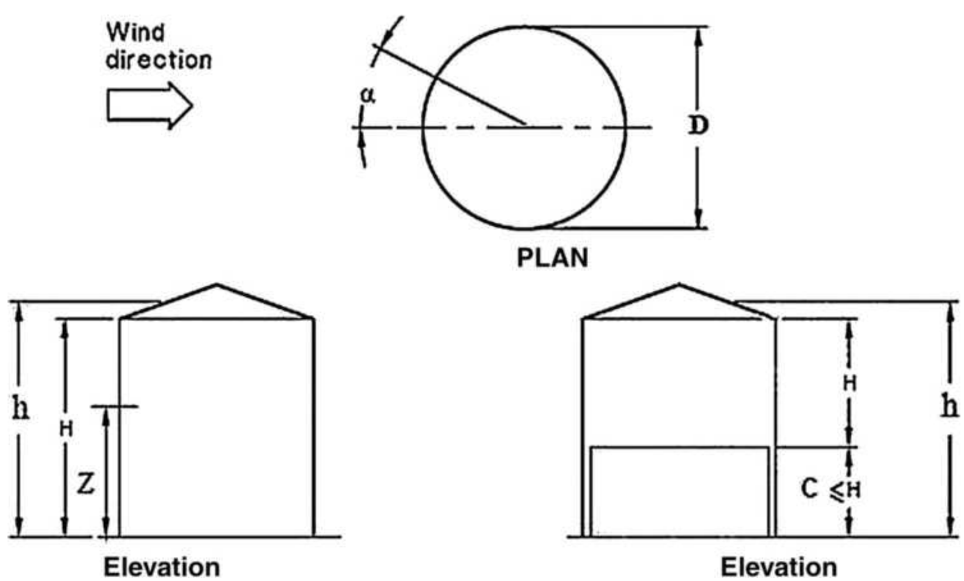
4. Wind forces shall be applied in the directions resulting in maximum member forces and reactions. For towers with square cross sections, wind forces shall be multiplied by the following factor when the wind is directed along a tower diagonal:

$$1 + 0.75\varepsilon, \text{but not} > 1.2$$

5. Wind forces on tower appurtenances such as ladders, conduits, lights, and elevators, shall be calculated using appropriate force coefficients for these elements.
6. Loads caused by ice accretion as described in Chapter 10 shall be accounted for.

FIGURE 29.4-3 Other Structures (All Heights): Force Coefficients, C_f, for Open Structures—Trussed Towers

Diagrams



Notation

C = Clearance height above the ground, in ft (m).
D = Diameter of a circular structure, in ft (m).
h = Mean roof height, in ft (m).
H = Solid cylinder height, in ft (m).
Z = Height to centroid of projected area of circular structure, in ft (m).
α = Angle from wind direction to a point on the wall of a circular bin, silo, or tank, in degrees.

FIGURE 29.4-4 Other Structures, Design Wind Loads for Main Wind Force Resisting Systems [h < 120 ft (h < 36.6 m)]: Circular Bins, Silos, and Tanks on the Ground or Supported by Columns, where $D \leq 120$ ft ($D \leq 36.6$ m), $0.25 \leq H/D < 4.0$

29.4.2.2 Roofs of Isolated Circular Bins, Silos, and Tanks. The net design pressures on the roofs of circular bins, silos, and tanks shall be determined from Eq. (29.4-4):

$$p = q_h(GC_p - (GC_{pi}))(\text{lb/ft}^2) \quad (29.4\text{-}4)$$

$$p = q_h(GC_p - (GC_{pi}))(\text{N/m}^2) \quad (29.4\text{-}4.\text{si})$$

where

q_h = velocity pressure for all surfaces evaluated at mean roof height h;
C_p = external pressure coefficient from Fig. 29.4-5 for roofs;
(GC_{pi}) = internal pressure coefficient for roofed structures from Section 26.13, and
G = gust-effect factor from Section 26.11.

The external pressures on the conical, flat, or dome roofs (roof angle less than 10°) of circular bins, silos, or tanks shall be equal to the external pressure coefficients, C_p, given in Fig. 29.4-5 for Zones 1 and 2. The external pressures for dome roofs (roof angle greater than 10°) shall be determined from Fig. 27.3-2.

29.4.2.3 Undersides of Isolated Elevated Circular Bins, Silos, and Tanks. External pressure coefficients C_p for the underside of elevated circular bins, silos, or tanks with clearance height, C, above ground less than or equal to the solid cylinder height, H, shall be taken as 0.8 and −0.6. For structures with clearance height above ground of less than or equal to one-third of the cylinder height, use linear interpolation between these values and $C_p = 0.0$ according to the ratio of C/h, where C and h are defined as shown in Fig. 29.4-4.

29.4.2.4 Roofs and Walls of Grouped Circular Bins, Silos, and Tanks. For closely spaced groups of three or more circular bins, silos, or tanks with center-to-center spacing less than $1.25D$, roof pressure coefficients, C_p, and drag force coefficient, C_f, on projected walls shall be calculated using Fig. 29.4-6. The net design pressures on the roofs shall be determined from Eq. (29.4-4). The overall drag shall be calculated based on Eq. (29.4-1).

29.4.3 Rooftop Solar Panels for Buildings of All Heights with Flat Roofs or Gable or Hip Roofs with Slopes Less Than 7°. As illustrated in Fig. 29.4-7, the design wind pressure for rooftop solar panels apply to those located on enclosed or partially enclosed buildings of all heights with flat roofs, or with gable or hip roof slopes with $\theta \le 7°$, with panels conforming to:

$L_p \le 6.7$ ft (2.04 m),
$\omega \le 35°$,
$h_1 \le 2$ ft (0.61 m),
$h_2 \le 4$ ft (1.22 m),

with a minimum gap of 0.25 in. (6.4 mm) provided between all panels, and the spacing of gaps between panels not exceeding 6.7 ft (2.04 m). In addition, the minimum horizontal clear distance between the panels and the edge of the roof shall be the larger of $2(h_2 - h_{pt})$ and 4 ft (1.2m) for the design pressures in this section to apply. The design wind pressure for rooftop solar panels shall be determined by Eq. (29.4-5) and (29.4-6):

$$p = q_h(GC_{rn})(\text{lb/ft}^2) \quad (29.4\text{-}5)$$

$$p = q_h(GC_{rn})(\text{N/m}^2) \quad (29.4\text{-}5.\text{si})$$

where

$$(GC_{rn}) = (\gamma_p)(\gamma_c)(\gamma_E)(GC_{rn})_{\text{nom}} \quad (29.4\text{-}6)$$

where

$\gamma_p = \min(1.2, 0.9 + h_{pt}/h)$;
$\gamma_c = \max(0.6 + 0.06L_p, 0.8)$; and
$\gamma_E = 1.5$ for uplift loads on panels that are exposed and within a distance $1.5(L_p)$ from the end of a row at an exposed edge of the array; $\gamma_E = 1.0$ elsewhere for uplift loads and for all downward loads, as illustrated in Fig. 29.4-7. A panel is defined as exposed if d_1 to the roof edge $> 0.5h$ and one of the following applies:

1. d_1 to the adjacent array $> \max(4h_2$, 4 ft (1.2m) or
2. d_2 to the next adjacent panel $> \max(4h_2$, 4 ft (1.2m).

$(GC_{rn})_{\text{nom}}$ = nominal net pressure coefficient for rooftop solar panels as determined from Fig. 29.4-7.

When, $\omega \le 2°$, $h_2 \le 0.83$ ft (0.25 m), and a minimum gap of 0.25 in. (6.4 mm) is provided between all panels, and the spacing of gaps between panels does not exceed 6.7 ft (2.04 m), the procedure of Section 29.4.4 shall be permitted.

The roof shall be designed for both of the following:

1. The case where solar collectors are present. Wind loads acting on solar collectors in accordance with this section shall be applied simultaneously with roof wind loads specified in other sections acting on areas of the roof not covered by the plan projection of solar collectors. For this case, roof wind loads specified in other sections need not be applied on areas of the roof covered by the plan projection of solar collectors.
2. Cases where the solar arrays have been removed.

29.4.4 Rooftop Solar Panels Parallel to the Roof Surface on Buildings of All Heights and Roof Slopes. The design wind pressures for rooftop solar panels located on enclosed or partially enclosed buildings of all heights, with panels parallel to the roof surface, with a tolerance of 2° and with a maximum height above the roof surface, h_2, not exceeding 10 in. (0.25 m) shall be determined in accordance with this section. A minimum gap of 0.25 in. (6.4 mm) shall be provided between all panels, with the spacing of gaps between panels not exceeding 6.7 ft (2.04 m). In addition, the array shall be located at least $2h_2$ from the roof edge, a gable ridge, or a hip ridge. The design wind pressure for rooftop solar collectors shall be determined by Eq. (29.4-7):

$$p = q_h(GC_p)(\gamma_E)(\gamma_a)(\text{lb/ft}^2) \quad (29.4\text{-}7)$$

$$p = q_h(GC_p)(\gamma_E)(\gamma_a)(\text{N/m}^2) \quad (29.4\text{-}7.\text{si})$$

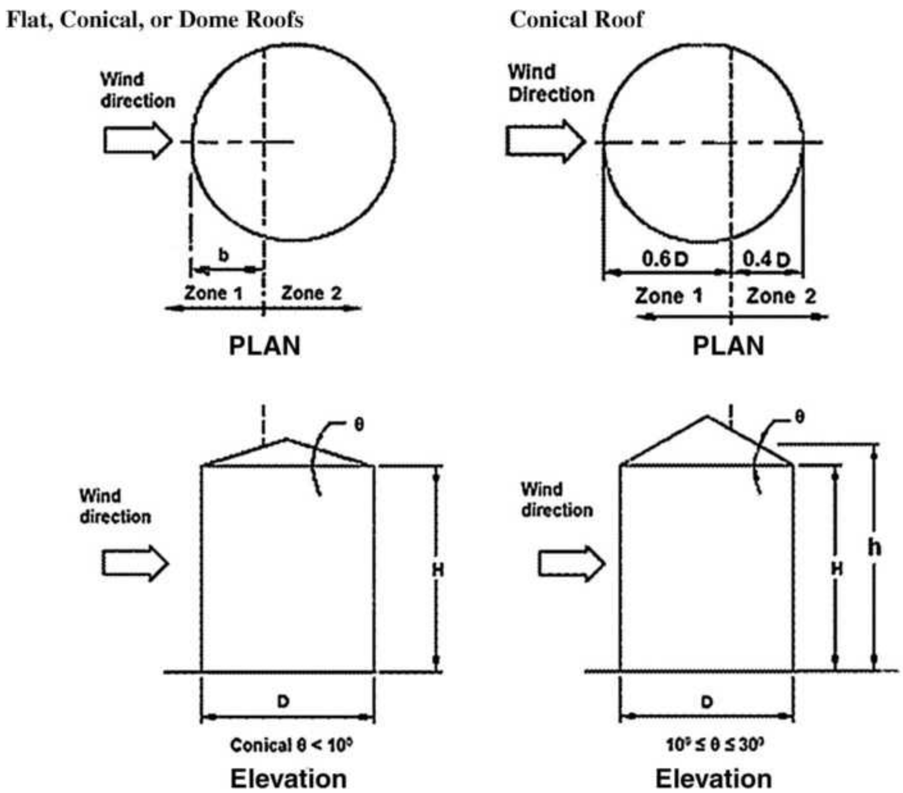
where

(GC_p) = external pressure coefficient for C&C of roofs with respective roof zoning, determined from Figs. 30.3-2A-I through 30.3-7 or 30.5-1;
γ_E = array edge factor = 1.5 for uplift loads on panels that are exposed and within a distance $1.5(L_p)$ from the end of a row at an exposed edge of the array; $\gamma_E = 1.0$ elsewhere for uplift loads and for all downward loads, as illustrated in Fig. 29.4-7. A panel is defined as exposed if d_1 to the roof edge $> 0.5h$ and one of the following applies:

1. d_1 to the adjacent array > 4 ft (1.2 m) or
2. d_2 to the next adjacent panel > 4 ft. (1.2 m);

γ_a = solar panel pressure equalization factor, defined in Fig. 29.4-8.

Diagrams



Notation

b = Determined below, in ft (m), dependent on H/D for roofs with average θ less than 10 degrees.
h = Mean roof height, in ft (m).
H = The solid cylinder height, in ft (m).
D = Diameter of a circular structure, in ft (m).
θ = Angle of plane of roof from horizontal, in degrees.

External Pressure Coefficients, C_p

Zone 1	−0.8
Zone 2	−0.5

Notes

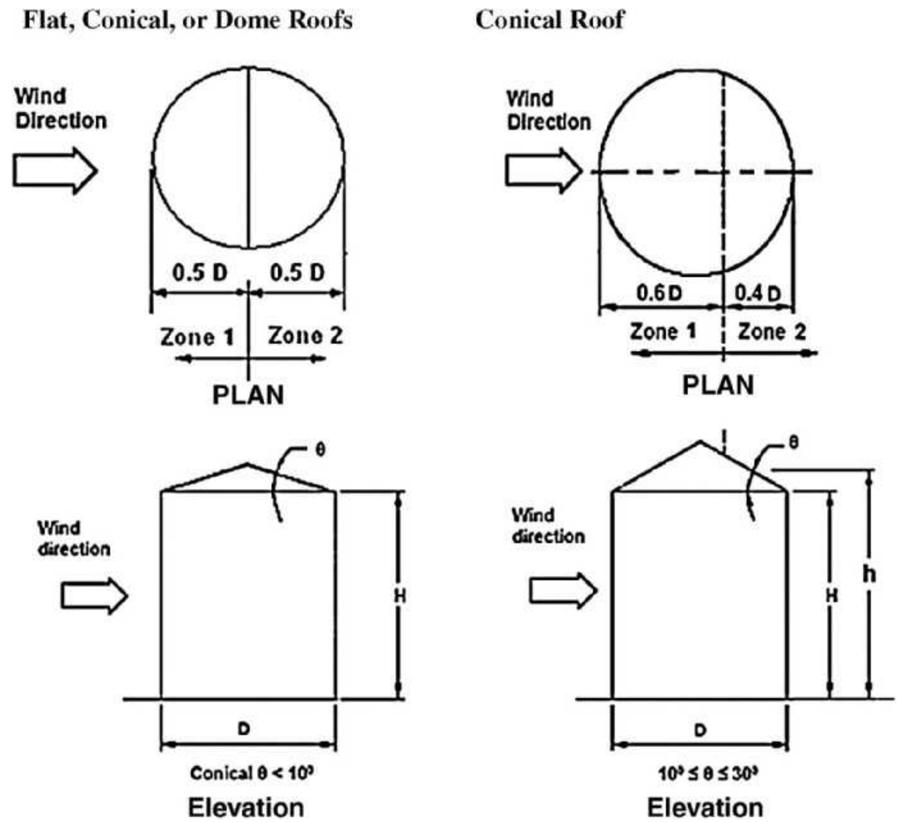
For roofs with average θ less than 10 degrees, the dimension, b, shall be determined as follows:

H/D	*b*
0.25	0.2D
0.5	0.5D
≥1.0	0.1h + 0.6D

Linear interpolation shall be permitted.

FIGURE 29.4-5 Other Structures, Design Wind Loads for Main Wind Force Resisting Systems [$h < 120$ ft ($h < 36.6$ m)]: External Pressure Coefficients, C_p, for Isolated Roofs of Circular Bins, Silos, and Tanks, where $D \leq 120$ ft ($D \leq 36.6$ m), $0.25 \leq H/D < 4.0$

Diagrams



Notation

D = Diameter of a circular structure, in ft (m).
h = Mean roof height, in ft (m).
H = The solid cylinder height, in ft (m).
θ = Angle of plane of roof from horizontal, in degrees.

Drag Force Coefficient (C_f) on Projected Walls

H/D	C_f	Use with
<1	1.3	q_h
2	1.1	q_h
4	1.0	q_h

Roof Pressure Coefficients, C_p, for Use with q_h

	H/D	Zone 1	Zone 2
$\theta < 10°$	≤ 0.5	−0.9	−0.5
	≥ 1.0	−1.3	−0.7
$10° < \theta < 30°$	≤ 4	−1.0	−0.6

FIGURE 29.4-6 Other Structures, Design Loads for Main Wind Force Resisting Systems [$h < 120$ ft ($h < 36.6$ m)]: Drag Force Coefficients, C_f, and Roof Pressure Coefficients, C_p, for Grouped Circular Bins, Silos, and Tanks on the Ground or Supported by Columns, Where $D \le 120$ ft ($D \le 36.6$ m), $0.25 \le H/D < 4.0$, and Center-to-Center Spacing ≤ 1.25

Diagrams

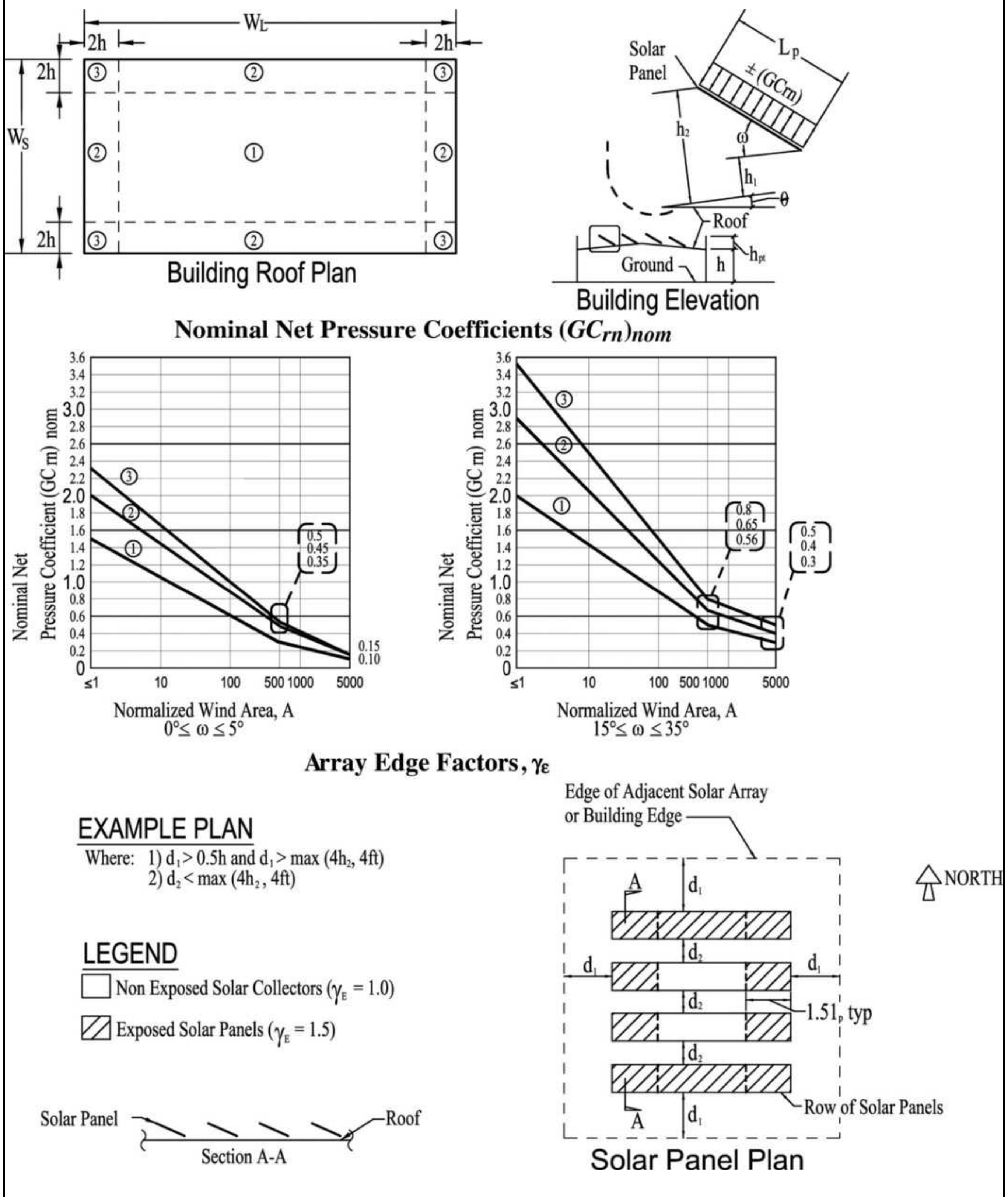


FIGURE 29.4-7 Design Wind Loads (All Heights): Rooftop Solar Panels for Enclosed and Partially Enclosed Buildings, Roof $\theta \leq 7°$

continues

Notation

A = Effective wind area, in ft^2 (m^2).
A_n = Normalized wind area, non-dimensional.
d_1 = For rooftop solar array, horizontal distance orthogonal to the panel edge to an adjacent panel or the building edge, ignoring any rooftop equipment in Fig. 29.4-7, in ft (m).
d_2 = For rooftop solar arrays, horizontal distance from the edge of one panel to the nearest edge in the next row in Fig. 29.4-7, in ft (m).
h = Mean roof height of a building except that eave height shall be used for roof angle θ less than or equal to 10°, in ft (m).
h_1 = Height of the gap between the panels and the roof surface, in ft (m).
h_2 = Height of a solar panel above the roof at the upper edge of the panel, in ft (m).
h_{pt} = Mean parapet height above the adjacent roof surface for use with Eq. (29.4-5), in ft (m).
L_p = Panel chord length.
W_L = Width of a building on its longest side in Fig. 29.4-7, in ft (m).
W_S = Width of a building on its shortest side in Fig. 29.4-7, in ft (m).
γ_E = Array edge factor as defined in Section 29.4.4.
θ = Angle of plane of roof from horizontal, in degrees.
ω = Angle that the solar panel makes with the roof surface in Fig. 29.4-7, in degrees.

Notes

1. (GC_{rn}) acts toward (+) and away (-) from the top surface of the panels.
2. Linear interpolation is allowed for ω between 5° and 15°.
3. $A_n = (1,000/[\max(L_b, 15)^2]A$, where A is the effective wind area of the structural element of the solar panel being considered, and L_b is the minimum of $0.4(hW_L)^{0.5}$ or h or W_s in ft (m).

FIGURE 29.4-7 (*Continued*). Design Wind Loads (All Heights): Rooftop Solar Panels for Enclosed and Partially Enclosed Buildings, Roof θ ≤ 7°

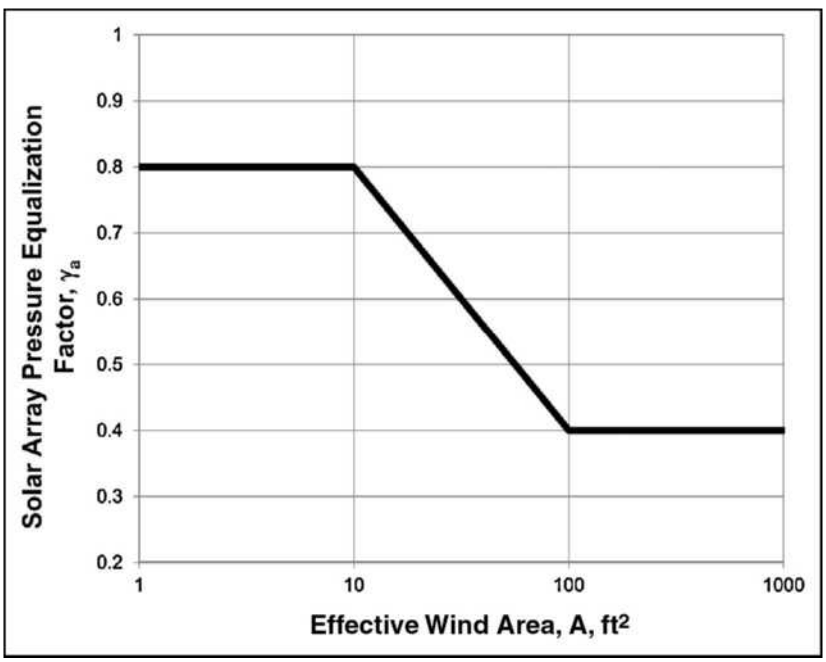


FIGURE 29.4-8 Solar Panel Pressure Equalization Factor, γ_a, for Enclosed and Partially Enclosed Buildings of All Heights

The roof shall be designed for both of the following:

1. The case where solar panels are present. Wind loads acting on solar collectors in accordance with this section shall be applied simultaneously with roof wind loads specified in other sections acting on areas of the roof not covered by the plan projection of solar collectors. For this case, roof wind loads specified in other sections need not be applied on areas of the roof covered by the plan projection of solar collectors.
2. Case where the solar panels have been removed.

29.5 PARAPETS

Wind loads on parapets are specified in Section 27.3.5 for buildings of all heights designed using the Directional Procedure and in Section 28.3.2 for low-rise buildings designed using the Envelope Procedure.

29.6 ROOF OVERHANGS

Wind loads on roof overhangs are specified in Section 27.3.4 for buildings of all heights designed using the Directional Procedure and in Section 28.3.3 for low-rise buildings designed using the Envelope Procedure.

29.7 MINIMUM DESIGN WIND LOADING

The design wind force for other structures shall be not less than 16 lb/ft^2 (0.77 kN/m^2) multiplied by the area A_f.

29.8 CONSENSUS STANDARDS AND OTHER REFERENCED DOCUMENTS

No consensus standards and other documents that shall be considered part of this standard are referenced in this chapter.